Where Do Semantics Live? A Homonym-Based Study of Semantic Disambiguation in Large Language Models

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***Abstract*—Transformer depth is sometimes interpreted as a processing hierarchy, raising the expectation that contextual meaning should become most clearly expressed within a reproducible semantic layer. We test this hypothesis using seven two-sense homonyms in four bidirectional encoders and four causal decoders, while separating three commonly conflated dimensions: layer depth, token position, and incrementally available context. Sense-conditioned representations are evaluated using held-out nearest-centroid margins and the Generalized Discrimination Value (GDV).**

Contextual sense was widely decodable, but its strongest expression did not concentrate at one stable depth. Profiling-selected layers achieved 0.917 held-out adequacy, compared with 0.893 at the final layer, although the crossed model–word interval included zero and effects varied substantially across models and homonyms. Adequacy-based cross-word selection likewise achieved higher point adequacy than GDV-based selection (0.914 versus 0.892), while the two criteria frequently preferred different depths. Global class separation and reliable item-level nearest-centroid discrimination therefore captured different properties of the activation geometry.

Context-position analyses further showed that layer depth cannot be interpreted independently of causal context availability. In causal decoders, homonym-position adequacy was constrained to 0.500 when the resolving context followed the homonym. At the sentence-final period, after that context had become available, adequacy reached 0.761, and 67.9% of initially inadequate observations became locally decodable.

Incremental context-conflict experiments showed broad updating but incomplete geometric resolution. Resolving information moved a fixed sentinel toward the authored sense in 77.2% of observations, yet only 40.3% of initially primed states crossed into the resolved-sense region. Conflict endpoints also remained 0.140 lower than matched non-conflicting endpoints containing the same homonym and resolver, indicating a persistent cost of preceding conflicting context.

The results do not support a single, objective-independent semantic stage. Contextual sense is better characterised as a distributed, readout-relative, and path-dependent property of transformer representations. Behavioural similarity may therefore motivate comparisons with human language processing without implying a shared ordering of intermediate operations.

***Index Terms*—large language models, semantic representations, homonyms, contextual updating, representational geometry, layer selection, encoders, causal decoders**

I. Introduction

Language models are judged at their outputs. Yet when their language behaviour resembles human behaviour, a harder question follows: do they reach that output through functionally analogous intermediate operations? Behavioural similarity motivates a search for functionally analogous intermediate computations, without implying mechanistic identity.

Human language processing offers an appealing, if simplified, ordering of such functions. Perceptual systems first transform an auditory or visual signal; lexical, syntactic, and semantic information is then integrated; task demands shape a response; and language or motor production supplies the observable endpoint. This ordering is functional rather than strictly feed-forward: human comprehension also involves incremental prediction and interactions between bottom-up input and top-down contextual expectations [1]. The beginning and end of this sequence are comparatively clear, while the intermediate operations overlap in time and remain harder to separate. Measures such as the N400 make part of this hidden processing visible by showing a systematic response when incoming material conflicts with semantic expectations. Semantic selectivity is also spatially structured across the cortex [2], and model predictability has been related to N400 and anticipatory neural activity [3]. These findings do not identify one neural semantic stage, but they motivate the search for functional signals between input and output.

A transformer invites an analogous hypothesis. It begins with an embedding representation, applies a sequence of increasingly composed transformations, and ends in an output distribution. Greater depth permits functions that aggregate information across more preceding computations, so it is tempting to expect an ordering from surface-form processing through syntax and semantics toward task- or output-relevant representations. A clean processing hierarchy is therefore an empirical hypothesis, not a consequence of the architecture.

Evidence from mechanistic interpretability complicates a clean hierarchy. Monosemantic and sparse-feature studies find concept-related directions at multiple depths [4–6]. That need not contradict a layer-level semantic signal: distributed features could still have their strongest combined, globally decodable expression within a narrower layer range. Feature location and population-level expression concern different scales of organisation.

The broad aim of this study is to test one tractable part of that proposed hierarchy: whether a layer-level signal of contextual semantic processing can be identified. The design uses homonyms because the same target form can support sharply different meanings. If a representation primarily preserves target form or grammatical role, two occurrences of *bank* should remain comparatively similar across its financial and riverside uses. As contextual sense becomes more strongly expressed, the two labelled groups should become more geometrically distinguishable. Contextual conflict can then ask whether that expression changes when later information contradicts an earlier interpretation.

This semantic assay is a necessary baseline rather than the complete hierarchy test. If a stable internal sense signal cannot first be identified, or if layer depth is confused with context availability or reading time, a broader comparison of syntactic, semantic, task, and neural processing stages would be premature. Homonyms hold the target form constant, but not the entire sentence: the authored contexts still differ in vocabulary, topic, construction, length, and target position. The dependent measure is therefore *sense-conditioned contextual separability*, not a pure semantic feature.

The study develops this baseline through six linked analyses, referred to throughout as H0–H5. H0 establishes the baseline geometric lean of bare and neutrally framed homonyms. H1 and H2 ask whether sense readout concentrates at particular layers and whether geometric layer selection transfers across words. H3 and H4 separate layer depth from causal context availability and local token-position readout. H5 then asks how a fixed readout changes when progressively longer, conflicting context is supplied. The following subsections motivate these analyses in that order before their formal methods are introduced.

A. Locating usable sense information

Previous work has shown that linguistic information is distributed non-uniformly across transformer depth, with different layers supporting different linguistic analyses [7]. Mechanistic interpretability research asks a finer-grained question by attempting to identify individual features or directions inside a layer. Superposition complicates that task because one unit can participate in several unrelated features [4]; sparse autoencoders address this by learning an overcomplete set of sparsely active directions [5, 6]. The present study does not attempt that form of feature recovery. It treats each layer as a candidate readout space and asks whether the two sense-conditioned groups occupy distinguishable regions there.

Two complementary quantities are used. A nearest-centroid margin asks whether the correct sense is closer under held-out evaluation. The Generalized Discrimination Value (GDV) directly summarises labelled within-sense and between-sense geometry without fitting a classifier [8]. It therefore provides a complementary criterion to the held-out nearest-centroid margin. Comparing them is informative precisely because agreement is an empirical question rather than an assumption. The limitations of flexible probing [9] motivate the geometric comparison, while nested held-out evaluation prevents the selected layer from being judged on the same sentences that selected it.

H1 and H2 ask whether contextual meaning becomes most clearly expressed at a reproducible stage of model depth. They approach this question through two different geometric definitions of sense separation. If both identify the same layer, or at least the same region of depth, this would support the broader hypothesis that semantic processing occupies a relatively stable position within an emerging order of aggregation and abstraction. If they disagree systematically, the idea of a single semantic stage becomes harder to sustain: where semantics appears to "live" would depend on which aspect of representational geometry is measured. The comparison therefore tests not merely which method selects a better layer, but whether semantic localisation is robust enough to support the larger hypothesis of an ordered processing hierarchy.

B. Three axes that must remain distinct

Transformer depth is sometimes described as though it were reading time, but the two are not interchangeable. This study separates three axes:

1) *Layer depth* indexes successive transformations within one forward pass.
2) *Sequence position* determines which context is causally available at a particular token.
3) *Incremental input* is studied by rerunning the model on progressively longer prefixes.

Both sampled model families build on multi-head self-attention [10], but differ in which context is available at a token position. Encoders in the BERT tradition use bidirectional attention [11], so the homonym can use both left and right context from the first attention layer. A causal decoder can use preceding context at every layer but can never use later words at the homonym position. Additional layers do not make right context become available retroactively. Conversely, a later token in the same decoder pass can attend to both the homonym and a later disambiguating cue.

H3 and H4 ask how the apparent location of semantic information depends on where context can be integrated and where the representation is read. This is necessary because a change across layers should not be confused with a change in the information available to the model. By comparing contexts that disambiguate the homonym before or after it, H3 tests whether architectural access to surrounding context determines when sense information can emerge. A strong encoder–decoder difference would show that semantic separability cannot be interpreted from layer depth alone: it is constrained by the model's causal structure.

H4 asks whether a sentence-level interpretation that is weak or unavailable at the homonym can nevertheless become accessible at a later token position. If so, semantic information is not attached exclusively to the ambiguous word, but can be expressed elsewhere as the sentence representation develops. If later positions do not recover the distinction, this would instead suggest a more persistent limitation in integrating the disambiguating context.

C. Baseline lean and incremental context conflict

A context-free or neutrally framed homonym can lie closer to one profiling-sense region than the other. H0 asks whether an ambiguous homonym already occupies an asymmetric position between the two contextual sense regions before either interpretation is explicitly selected. Human comprehenders interpret language incrementally and can experience garden-path costs when later material conflicts with an earlier interpretation [12] [13]. That literature motivates asking whether a model-internal readout changes when conflicting context is extended, but the present experiment does not measure human ambiguity, reading time, surprise, or reanalysis. The stimuli are therefore called authored *context-conflict items*.

H5 extends the question from semantic localisation to semantic revision. A useful representation of meaning must not only distinguish between senses when the context is clear; it must also change when new information overturns an earlier interpretation. Temporarily conflicting homonym contexts therefore provide a controlled test of whether the model updates its internal sense representation or remains biased by the interpretation established first.

Successful revision would indicate that sense-conditioned representations remain flexible as context develops. Incomplete revision would instead reveal path dependence: later evidence influences the representation but does not fully overcome its earlier semantic state. This matters beyond lexical ambiguity, because the same capacity is required whenever a model must revise an interpretation, belief, or represented state of the world in response to new information. H5 therefore asks not where semantic information is most separable, but how readily that information can be updated over the course of incremental context. It provides a model-internal analogue of semantic revision, without claiming that the process is equivalent to human garden-path reanalysis.

D. Research questions and operational expectations

The six analyses are stated in terms of the quantities they can identify:

- H0 Do bare and neutrally framed homonym states show a consistent directional lean toward one of the two profiling-sense regions at the selected contextual layer?
- H1 Does nested held-out adequacy at a profiling-selected contextual layer exceed the final-layer default, and how heterogeneous is selected depth across model–word pairs?
- H2 When transferred to a held-out homonym, how do GDV-based and adequacy-based cross-word layer selection compare with the final layer and a held-out-word oracle in adequacy, oracle regret, and layer distance?
- H3 Does causal availability constrain homonym-position sense decodability, as indicated by a larger paired left-minus-right margin difference in the sampled causal decoders than in the sampled encoders?
- H4 In causal decoders, how often does an authored sentence sense that is not locally decodable at the homonym become decodable at the sentence-final period after the resolving context has been encountered, and how do the corresponding gain and loss rates compare with bidirectional encoders?
- H5 Does the fixed-sentinel margin move toward the authored resolved-sense region after the resolver is added; how often does it cross the centroid boundary among initially primed observations; and how do conflict endpoints compare with matched non-conflicting and resolver-only endpoints?

Together, H0–H5 test whether contextual semantics occupies a reproducible stage of transformer depth and whether the resulting sense representation remains flexible as context changes. Convergence across words, models, and geometric criteria would support a stable semantic stage; dependence on readout position, metric, or contextual history would instead indicate a more distributed and path-dependent organisation.

II. Data and Methods

The study uses the geometry of contextual hidden states to ask three distinct questions. First, *layer depth* compares successive transformations within one forward pass. Second, *sequence position* compares representations at different token positions when the same completed sentence is supplied to the model. Third, *incremental context* reruns the model on progressively longer prefixes and therefore changes what input is available between forward passes. These axes are analysed separately: H1–H2 concern layer depth, H3–H4 concern sequence position, and H5 concerns incremental context.

A. Shared design

a) Operational sense contrasts.: Seven English homonyms were assigned two operational sense labels (Table I). The labels define the binary contrasts evaluated in this study; they are not intended as exhaustive lexical inventories. In particular, words such as *spring* and *pitch* have additional readings that are outside the analysis. An eighth candidate homonym, *light*, was also authored but excluded from the study. Since its two senses (illumination noun vs. low-weight adjective) differ in part of speech, sense ambiguity cannot reliably be engineered into realistic sentences, which introduces a confound for every hypothesis in which ambiguity and its resolution are essential (H3–H5).

All stimuli were authored for this model-internal study. Holding the target word constant removes lexical identity as an explanation for the sense contrast, but it does not isolate semantics from every other property of the sentence. The two sense sets also differ in contextual vocabulary, topic, syntax, and sometimes target position. The dependent measures are therefore described as *sense-conditioned contextual separability* or *local sense decodability*, not as a pure measure of semantics.

b) Profiling sentences.: The profiling set was authored with 20 unambiguous sentences per sense and word (280 sentences) (Table II). Activation extraction required a standalone occurrence of the target word, or its regular plural, located with word-boundary-aware matching. Where a target occurrence was split into multiple subword tokens, their hidden states were mean-pooled. For example, Qwen2.5 tokenised *Cranes* in “Cranes migrate thousands of miles . . . ” as `C`, `ran`, and `es`; the three corresponding hidden states were averaged to

TABLE I
THE SEVEN HOMONYMS AND THE TWO SENSE CONTRASTS ANALYSED.

Word	Sense 0	Sense 1
bank	river / embankment	financial institution
bark	dog vocalisation	tree covering
bat	sports equipment	flying mammal
crane	large wading bird	lifting machine
match	sports contest	fire-starting stick
spring	season	water source
pitch	sports field	business proposal

represent the target occurrence. These homonym-position states provide the reference centroids used throughout the study.

For H4, the same profiling pass also cached the state at the final non-special, non-padding token. In all profiling and H4 sentences this token is the sentence-final period. Period-position centroids therefore provide a common endpoint geometry for H4. This readout asks whether the sentence sense is locally decodable from a fixed punctuation token after the complete sentence has been processed; it does not treat the period as a meaning-bearing word or claim that the earlier homonym state has moved there.

TABLE II
SAMPLE PROFILING STIMULI FOR *bank* IN H1. THE COMPLETE SET CONTAINS 20 SENTENCES FOR EACH SENSE. IN EACH OUTER FOLD, ONE OF THE 40 SENTENCES IS HELD OUT, WHILE THE REMAINING 39 ARE USED TO SELECT A LAYER AND ESTIMATE THE SENSE CENTROIDS. EVERY SENTENCE SERVES AS THE HELD-OUT ITEM ONCE.

Sense	Operational label	Example profiling sentence
0	Riverside	There's a steep bank at the bend of the river.
0	Riverside	Fishermen sat quietly on the bank waiting for a bite.
1	Financial	She went to the bank to open a new account.
1	Financial	The bank manager approved her loan application.

c) Models.: The analysis includes four bidirectional encoders—ModernBERT-large, DeBERTa-v3-large, RoBERTa-large, and XLM-RoBERTa-large—and four causal decoders—Qwen2.5-3B, Qwen2.5-7B, Mistral-Nemo-Base-2407, and OLMo-2-1124-7B. All are open-weight base checkpoints, allowing hidden states to be extracted at every layer without instruction-tuning as an additional source of variation. The model sample is deliberately heterogeneous. Encoder and decoder groups also differ in parameter count, training objective, training data, tokenizer, and depth; consequently, broad group differences are descriptive unless the experimental manipulation itself identifies the role of causal availability. H3 provides that stronger comparison by presenting the same words in paired left- and right-context orders.

B. Representational measures

a) Adequacy margin.: For a hidden state h_l at layer l, let $c_{0,l}$ and $c_{1,l}$ be the profiling centroids for the two senses at the relevant readout position. For an observation whose authored sense label is $s \in \{0, 1\}$, the Euclidean adequacy margin is

$$M_l(h, s) = \|h_l - c_{1-s,l}\|_2 - \|h_l - c_{s,l}\|_2. \tag{1}$$

Thus $M_l > 0$ places the observation closer to its labelled sense centroid, $M_l < 0$ places it closer to the other centroid, and $M_l = 0$ lies on their nearest-centroid decision boundary. The correct and incorrect centroids are assigned separately for every observation. This symmetric scoring is essential for balanced datasets: using one fixed centroid direction would reverse the intended sign for half of the sentences.

To compare magnitudes across layers and architectures, the raw margin is normalised by the inter-centroid distance,

$$\widehat{M}_l(h, s) = \frac{M_l(h, s)}{\|c_{1,l} - c_{0,l}\|_2}. \tag{2}$$

By the reverse triangle inequality, this normalised Euclidean margin is bounded exactly by $-1 \leq \widehat{M}_l \leq 1$. The two centroids attain the endpoint values, and zero denotes the nearest-centroid decision boundary. Normalisation removes the direct scale dependence of the raw distance difference, which is important when comparing models and layers with different activation magnitudes. Unless stated otherwise, binary adequacy uses the unthresholded sign ($M_l > 0$), and magnitude comparisons use $\widehat{M}_l$.

b) Centroid estimation and layer use.: When profiling sentences are themselves evaluated in H1 and H2, each sentence is scored against centroids estimated without that sentence. H1 additionally nests layer selection inside an outer leave-one-sentence-out fold. For H0 and H3–H5, centroids and the selected layer are estimated from the separate profiling set before the hypothesis-specific stimuli are scored. This prevents the tested carrier, word-order, or incremental-context item from contributing to its own reference centroid or layer choice.

c) Uncertainty and thresholds.: Model–word cells, rather than repeated sentence rows, are the evidential units for architecture-level summaries. Contrasts in H1–H3 are accompanied by 95% intervals from 20,000 crossed bootstrap resamples of models and words. H4 reports pooled counts and proportions for its conditional transitions so that the contributing denominators remain visible. Cutoffs such as $|\widehat{M}_l| > 0.3$ are used only as descriptive sensitivity summaries; continuous margins and unthresholded signs remain the primary outcomes.

d) Exploratory analyses across hypotheses and architectures.: Two post-hoc analyses examined relationships not specified by the primary hypotheses. First, we asked whether the baseline sense lean measured in H0 was associated with H1 adequacy or H5 updating. The H0–H1 analysis used all 56 model–word combinations (8 models $\times$ 7 homonyms). For H5, results were additionally separated by the two conflict directions, producing 112 model–word–direction combinations. A transition rate could be calculated for 107 of these combinations; in the remaining five, no item was classified as primed at the homonym stage, leaving the conditional rate without a denominator. Spearman correlations were calculated across these aggregated combinations. Because the relationships were selected after inspection of the primary results, they are treated as descriptive and were not corrected for multiple comparisons.

Second, we explored whether H5 outcomes differed between the sampled encoders and decoders. Outcomes were first averaged within each of the 56 model–word combinations. Decoder-minus-encoder differences and their 95% intervals were then estimated using 20,000 crossed bootstrap resamples of models and words. The 20,000 resamples quantify uncertainty and do not represent additional stimuli or model observations.

C. H0: baseline sense lean

H0 separates two choices that are easy to conflate: the *input supplied to the model* and the *layer from which its activation is read*. In the bare-word condition the input contains only the homonym (apart from any model-added special tokens) (Table III).

TABLE III
EXAMPLE H0 INPUTS FOR *bank*. THE BARE WORD AND ALL FIVE NEUTRAL CARRIERS CONTAIN NO CLAUSE THAT EXPLICITLY SELECTS EITHER THE RIVERSIDE OR FINANCIAL SENSE. EACH INPUT UNDERGOES A COMPLETE FORWARD PASS, AND THE HOMONYM STATE IS READ AT THE INDEPENDENTLY SELECTED CONTEXTUAL LAYER.

Condition	Model input
Bare word	`bank`
Neutral carrier 1	The bank was unstable.
Neutral carrier 2	They examined the bank.
Neutral carrier 3	The bank had changed.
Neutral carrier 4	The bank needed protection.
Neutral carrier 5	The bank became difficult to access.

In the carrier condition it contains the homonym inside a neutral, ambiguous sentence. Neither input contains context selecting one of the two senses. Both inputs nevertheless undergo a complete forward pass through the transformer:

$$\text{bare input } \texttt{bank} \longrightarrow h_0 \longrightarrow h_1 \longrightarrow \cdots \longrightarrow h_{l^*}. \tag{3}$$

Here h_0 is the pre-transformer embedding output and h_{l^*} is the homonym state at the independently selected layer for that model–word pair. Thus *context-free* describes the input; it does not mean that the activation is taken from the embedding layer.

The selected layer is used because H0 asks where the bare or neutrally framed word lies relative to the two sense regions used by the rest of the study. Those regions are defined by the two profiling centroids at the same selected layer. At layer 0, by contrast, the intended river and financial readings of *bank* receive the same lexical embedding when token form and position are held constant: no surrounding words have yet been integrated. Any apparent layer-0 difference between the authored sense groups can therefore arise from position, capitalisation, inflection, or tokenisation rather than semantic disambiguation. H1 reports layer 0 separately as a pre-attention negative control.

Accordingly, H0 estimates a *baseline sense lean in the selected contextual geometry*. It is neither an embedding-layer analysis nor an estimate of which sense is more frequent in pretraining; no corpus-frequency measure is included in the study.

For each word, the bare word was presented once and five ambiguous carrier sentences were presented without either disambiguating clause. All states were read at the homonym position and at the layer selected from the independent profiling set. A signed coordinate was defined as

$$B_l(h) = \frac{\|h_l - c_{1,l}\|_2 - \|h_l - c_{0,l}\|_2}{\|c_{1,l} - c_{0,l}\|_2}, \tag{4}$$

where positive values lean toward sense 0 and negative values toward sense 1. The same carrier had previously been scored once under each opposing "correct" label, producing the redundant pair M and $-M$. These mirrors were collapsed into one signed observation, leaving five independent carrier states per model–word combination rather than ten. H0 reports the bare-word lean, the carrier lean and its absolute magnitude, the shift from the bare word to the carrier, and the proportion of carriers agreeing in direction. The count with $|B_l| > 0.3$ is a descriptive sensitivity summary only.

D. H1: layer depth and held-out adequacy

H1 asks whether choosing a contextual layer from profiling data improves sense-conditioned separability relative to taking the final layer by default. At every contextual layer (the embedding output, layer 0, was not a selection candidate), profiling sentences were scored with leave-one-sentence-out sense centroids.

Layer 0 was nevertheless retained as a descriptive negative control. It is the pre-transformer embedding output and therefore precedes any self-attention. Apparent label separation at this layer cannot be attributed to contextual sense integration; it instead diagnoses differences such as target position or surface-token form between the two authored sense sets. Selection begins at layer 1 because H1 asks which *contextual* transformation is most useful.

Performance was estimated with a nested leave-one-sentence-out procedure. In each outer fold, one profiling sentence was held out. On the remaining sentences, the layer with the highest inner leave-one-out fraction adequate was selected, with mean normalised margin breaking ties. The untouched sentence was then scored both at that selected layer and at the final layer using centroids estimated only from the outer training sentences. This yields a paired selected-minus-final outcome without evaluating a sentence at a layer chosen using that sentence. A separate best layer estimated from the complete profiling set with leave-one-out scoring was retained only for the genuinely separate H0 and H3–H5 stimulus sets. The main H1 contrast is the crossed model–word mean of nested selected-layer minus final-layer adequacy; modal selected layer and relative depth describe the heterogeneity of layer choice.

E. H2: cross-word layer-selection strategies

H2 asks whether a layer choice generalises to a new homonym. Each word was held out in turn, and layer selection used only the other six words. The held-out word was then evaluated with its own leave-one-sentence-out centroids. Four strategies were compared:

1) *GDV*: the layer with the most negative mean GDV across the six profiling words;
2) *adequacy-based cross-word*: the layer with the highest mean leave-one-out adequacy across those words;
3) *last*: the final layer, requiring no selection; and
4) *oracle*: the best layer on the held-out word itself, included only as an optimistic ceiling and not as a deployable strategy.

For a layer containing D-dimensional states, features were standardised as $x'_{nd} = \frac{1}{2}(x_{nd} - \mu_d)/\sigma_d$, following the published GDV definition. GDV was then computed as

$$\text{GDV} = \frac{\bar{d}_{\text{intra}} - \bar{d}_{\text{inter}}}{\sqrt{D}}, \tag{5}$$

where $\bar{d}_{\text{intra}}$ averages Euclidean distances within the two sense classes and $\bar{d}_{\text{inter}}$ averages distances between them. More negative values indicate stronger separation between the two labelled sense groups. Because GDV is calculated directly from their geometry, it does not require fitting a classifier.

Exact agreement with the oracle layer is too brittle to be the primary criterion because adjacent layers may perform almost identically. The strategy comparison instead reports held-out fraction adequate, regret relative to the oracle, layer distance from the oracle, and the proportion of selections within two layers of the oracle. Crossed model–word bootstrap intervals compare GDV-based selection, adequacy-based selection, and the oracle with the last-layer baseline.

F. H3: causal availability at the homonym position

H3 uses paired sentences to manipulate whether decisive context is available at the homonym position. Each word has five ambiguous carriers. Every carrier is combined with one clause for each sense and appears in two orders: condition L places the clause before the carrier, whereas condition R places it after the carrier (Table IV). The two orders contain the same lexical material. This gives $5 \times 2 \times 2 = 20$ sentences per word and 140 sentences overall.

TABLE IV

ONE OF THE FIVE H3 CARRIER SETS FOR *bank*. THE SHARED CARRIER IS SHOWN IN BOLD, WHILE THE DISAMBIGUATING CLAUSE VARIES BY SENSE AND APPEARS EITHER BEFORE (L) OR AFTER (R) THE HOMONYM. EACH CARRIER THEREFORE PRODUCES FOUR SENTENCES.

Condition	Intended sense	Sentence
L	Riverside	After weeks of heavy rain, **the bank was unstable**.
R	Riverside	**The bank was unstable** after weeks of heavy rain.
L	Financial	After weeks of market panic, **the bank was unstable**.
R	Financial	**The bank was unstable** after weeks of market panic.

Hidden states were extracted at the homonym at the profiling-selected layer and scored against homonym-position profiling centroids. Each L sentence was paired with its R reordering, and the primary outcome was the L-minus-R change in normalised margin. Effects were first aggregated within model–word combinations. The architecture contrast was the decoder-minus-encoder difference in this paired L–R effect, with a crossed model–word bootstrap interval. For causal decoders, the two R-condition sentences associated with each carrier have an identical prefix at the homonym but opposing sense labels. Decoder R adequacy of 0.500 is a construction-based manipulation check arising from the identical homonym-stage prefixes with opposing labels; the empirical quantity of interest is the magnitude of the paired L–R margin difference and its architecture interaction.

G. H4: sequence-position decodability

H4 uses the ten R-condition sentences per word and compares two readout positions from the same completed forward pass: the homonym and the final non-special, non-padding token. All 560 analysed observations end in a period, so the second readout is the sentence-final period. Holding the layer fixed is deliberate: H4 tests whether changing the readout position after additional context has become available makes the authored sense locally decodable, rather than allowing both token position and layer depth to vary.

At the selected layer, the homonym state was scored against profiling centroids estimated at the homonym position (*target-local*), while the period state was scored against profiling centroids estimated at the sentence-final period (*final-local*). This provides a position-appropriate measure of sense decodability at each readout. The period state was additionally scored against the homonym-position centroids to measure how strongly the representational geometry changes between the two positions.

The primary outcomes are derived from the target-local by final-local 2×2 table. The gain rate,

$$P(\text{final adequate} \mid \text{target inadequate}),$$

measures how often a sense that is not decodable at the homonym becomes decodable at the sentence endpoint. The loss rate,

$$P(\text{final inadequate} \mid \text{target adequate}),$$

measures the reverse transition. Pooled counts and proportions are reported for both conditional rates.

The endpoint provides a later sentence-level readout after the decisive context has been encountered. The comparison therefore captures the combined effect of accumulated context and readout position. Token identity, position, clause length, and the distance between the decisive cue and the endpoint vary within this comparison and are treated as properties of the later sentence representation.

H. H5: incremental context-conflict updating

H5 separates incremental input from both layer depth and sequence position by rerunning each model on progressively longer prefixes while holding the readout token constant.

TABLE V
EXAMPLE H5 INPUTS FOR ONE *bank* ITEM PRIMING THE FINANCIAL SENSE AND RESOLVING TO THE RIVERSIDE SENSE. THE MODEL IS RERUN FOR EACH ROW. WITHIN EACH INPUT CELL, THE INDENTED SENTINEL `PROBE` FORMS THE FINAL PARAGRAPH OF THAT SAME INPUT. THE MATCHED CONTROL CONTAINS THE SAME HOMONYM AND RESOLVER WITHOUT THE CONFLICTING FINANCIAL PRIME. THIS IS ONE OF 14 *bank* ITEMS; THE COMPLETE SET CONTAINS SEVEN ITEMS IN EACH CONFLICT DIRECTION.

Input condition	Function	Exact input supplied to the model
Prime stage	Financial prime only	She finished her loan application and walked to the **probe**
Homonym stage	Prime plus ambiguous word	She finished her loan application and walked to the bank **probe**
Resolution stage	Conflict through the resolver	She finished her loan application and walked to the bank to sit by the river **probe**
Matched control	Non-conflicting context through the same resolver	After following the riverside path, she walked to the bank to sit by the river **probe**
Resolver in isolation	Lexical resolver baseline	river **probe**

The final set contains 98 items across seven words, with 14 items for each word. Within every word, seven items prime sense 0 and resolve to sense 1, and seven prime sense 1 and resolve to sense 0. Thus words and conflict directions contribute equally to the pooled design. Every item has a matched non-conflicting control containing the same homonym and resolver, and all items passed a structural check that the order is prime, homonym, then resolver.

For each item, the neutral sentinel word `probe` was appended in a new paragraph to (i) the prime ending immediately before the homonym, (ii) the prefix through the homonym, and (iii) the prefix through the annotated resolver (Table V). The model was rerun for each prefix and the sentinel's state was extracted at the same profiling-selected layer. Appending the sentinel in a separate paragraph prevents it from completing a different local syntactic slot at different prefix lengths. Sense centroids for the sentinel were estimated from independent unambiguous profiling sentences with the same sentinel appended. Consequently, all three stages use the same sentinel token, terminal readout role, selected layer, and sense axis. The sentinel's absolute sequence position necessarily changes as the supplied prefix grows; the experimental manipulation is therefore the preceding context available at that terminal readout.

The primary continuous outcome is the paired change $\Delta\widehat{M} = \widehat{M}_{\text{resolution}} - \widehat{M}_{\text{homonym}}$ toward the authored correct sense. A positive value indicates updating in the intended direction but does not require a boundary crossing. The conditional transition rate is reported only among items whose homonym-stage sentinel first lies on the primed side: $P(\text{resolution correct} \mid \text{homonym primed})$. Two controls qualify the interpretation. First, the resolution-stage margin is compared with the matched non-conflicting sentence through the same resolver, estimating the cost of preceding conflict. Second, it is compared with the resolver presented in isolation before the sentinel, showing how much of the endpoint can be attributed to the resolver's lexical content alone. Thresholds of 0.1 and 0.3 are retained only as sensitivity analyses; the continuous paired change is the primary result.

III. RESULTS

A. H0: baseline sense lean

Ambiguous carriers showed substantial and directionally consistent lean toward one of the two analysed senses. Across 56 model–word combinations, the mean absolute normalised carrier lean was 0.339 and the mean directional consistency across the five carriers was 0.882. In 159 of 280 independent carrier states, $|B_l|$ exceeded the descriptive 0.3 cutoff. The corresponding values were 65/140 for encoders and 94/140 for decoders, with mean absolute lean of 0.279 and 0.398 respectively (Table VI).

TABLE VI
ARCHITECTURE-LEVEL SUMMARY OF H0 BASELINE SENSE LEAN IN NEUTRAL CARRIERS. MEAN $|B_l|$ IS THE ABSOLUTE NORMALISED DIFFERENCE IN DISTANCE TO THE TWO SENSE CENTROIDS; LARGER VALUES INDICATE A STRONGER RELATIVE LEAN TOWARD EITHER CENTROID. DIRECTION CONSISTENCY IS THE LARGER OF THE POSITIVE AND NEGATIVE PROPORTIONS AMONG NON-ZERO CARRIER LEANS. LARGER VALUES INDICATE STRONGER DIRECTIONAL AGREEMENT, AND 1.0 INDICATES THAT ALL NON-ZERO CARRIERS LEAN IN THE SAME DIRECTION. IF ALL FIVE LEANS ARE NUMERICALLY ZERO, CONSISTENCY IS ASSIGNED A VALUE OF 0.5. THE FINAL COLUMN COUNTS INDIVIDUAL CARRIER STATES WITH A COMPARATIVELY STRONG LEAN, $|B_l| > 0.3$, OUT OF 140 OBSERVATIONS PER ARCHITECTURE.

Architecture	Mean $\lvert B_l \rvert$	Direction consistency	$\lvert B_l \rvert > 0.3$
Encoder	0.279	0.864	65/140
Decoder	0.398	0.900	94/140
All models	0.339	0.882	159/280

The strongest baseline leans were semantically specific. Qwen2.5-3B leaned toward the tree-covering sense of *bark* ($B_l = -0.657$) and the water-source sense of *spring* ($B_l = -0.617$). OLMo-2-1124-7B likewise leaned toward the water-source sense of *spring* ($B_l = -0.594$). All three combinations had 100% cross-carrier directional consistency. These results establish a model-specific baseline lean within the study's two-sense geometry.

The word-level pattern was more systematic than the strongest individual model results alone suggest. Seven of eight models leaned toward the financial sense of *bank*, all eight toward tree covering for *bark*, seven toward the water-source sense of *spring*, and seven toward the sporting sense of *match*. In contrast, *bat* had an across-model mean close to zero and the lowest carrier-direction consistency of the seven words. Thus some homonyms show a broadly shared baseline direction across models, whereas *bat* provides no stable model-general prior in the present geometry (Figure 1).

Bare-word and carrier leans were not interchangeable. Across the 56 model–word combinations, their directions agreed in 28 cases. Agreement was strongest for *bank* (7/8 models) and *match* (5/8), whereas *spring* (2/8) frequently changed direction once placed in a carrier. For many model–word pairs, the mean carrier lean lay farther from zero than the bare-word lean, although carrier context could also weaken or reverse the bare-word direction. For example, Qwen2.5-7B changed from a strong bare-word bird lean for *crane* to a weaker but directionally consistent machine lean across the carriers. The isolated word therefore does not determine the contextual baseline: underspecified sentence context can amplify, attenuate, or reverse its position within the two-sense geometry.

B. H1: nested layer selection

Before contextual layer selection, the layer-0 negative control was almost exactly at chance overall: mean leave-one-out adequacy was 0.500 across the 56 model–word combinations. Word means nevertheless ranged from 0.238 for *bank* to 0.675 for *crane*. These deviations cannot reflect integration of sentence context because layer 0 precedes self-attention. The profiling sets also differ in target-position distributions between senses—for example, *bat* occurs after an average of 4.1 preceding words in sense 0 but 2.0 in sense 1, and *pitch* after 5.2 versus 2.45. Layer-0 performance is therefore treated as a stimulus-position and tokenisation diagnostic, not as evidence that a disambiguated sense is already represented in the embedding.

Across 2,240 outer folds, nested layer selection achieved 0.917 adequacy, compared with 0.893 at the final layer, a pooled difference of 2.37 percentage points (Table VIII). The point difference was small for encoders (0.63 points) and larger for decoders (4.11 points). However, the crossed model–word mean effect was 0.0237 with a 95% bootstrap interval of $[-0.0058, 0.0598]$, which includes zero.

The descriptive full-profile layer choices also differed by architecture. Encoder optima were concentrated relatively late in the network, with a mean relative depth of 0.762 and a median of 0.866; none of the 28 encoder–word combinations selected a layer in the first third of the network (one combination, DeBERTa-v3-large on *spring*, sat exactly at the one-third boundary). Decoder choices were earlier and more dispersed, with a mean relative depth of 0.494 and a median of 0.366; 13 of 28 decoder–word combinations selected a layer in the first third, while six selected the final layer. The encoder mean was later than the decoder mean for every homonym, although the model-specific ranges overlapped substantially. This is an architecture-level tendency rather than a shared semantic layer: individual choices still ranged from approximately one tenth of network depth to the final layer.

The average conceals substantial model heterogeneity (Figure 2). Selection improved Qwen2.5-3B by 9.29 points and Mistral-Nemo-Base-2407 by 6.79 points, but was slightly worse than the final layer for Qwen2.5-7B, ModernBERT-large, and RoBERTa-large. H1 therefore supports the practical claim that the final layer is not uniformly optimal, especially among the decoders sampled here, but it does not establish a model-independent average advantage for learned layer selection.

C. H2: cross-word selection and the GDV objective

GDV-selected layers achieved 0.892 mean held-out adequacy, essentially the same as the final-layer baseline of 0.893 (effect $= -0.0009$, 95% CI $[-0.0366, 0.0308]$). In contrast, adequacy-based cross-word selection achieved 0.914 (effect $= 0.0210$, 95% CI $[-0.0013, 0.0496]$). The held-out-word oracle reached 0.930 (effect relative to last $= 0.0371$, 95% CI $[0.0112, 0.0728]$), showing that useful layer variation existed even when GDV did not exploit it reliably (Table IX). Figure 3 provides a separate full-profile diagnostic of within-word agreement between the two objectives; it is not the leave-one-word-out H2 transfer result.

The two cross-word selectors rarely converged on the same depth. GDV-based and adequacy-based selection chose the identical layer in 7 of 56 folds (12.5%) and layers within two positions of one another in 14 folds (25.0%); their median absolute separation was four layers. The disagreement was therefore not limited to adjacent, practically equivalent choices.

The performance consequences were concentrated among the causal decoders. For encoders, GDV-based selection, adequacy-based selection, and the final layer all achieved similar adequacy (0.960, 0.966, and 0.960), close to the encoder oracle of 0.980. For decoders, adequacy-based selection reached 0.862, compared with 0.824 for GDV and 0.826 for the final layer, against an oracle of 0.879. Adequacy-based cross-word selection therefore recovered much of the available decoder improvement, whereas GDV did not.

The word-level results were also heterogeneous. *Crane* had the lowest adequacy under GDV-based selection (0.838), whereas *spring* (0.075) and *bark* (0.072) showed larger oracle regret than *crane* (0.038), making them the harder transfer cases by that measure. *Bank* was stable under either cross-word criterion (0.900 GDV versus 0.903 adequacy-based). *Match* showed the largest margin by which GDV-based selection exceeded adequacy-based selection on average (0.900 versus 0.881); *pitch* showed the same direction by a much smaller margin (0.888 versus 0.884). Although *bat* supplies the clearest individual geometric counterexample below, it was not a general cross-model failure: its mean adequacy was 0.925 under GDV-based selection and 0.959 under adequacy-based selection.

Layer proximity and predictive adequacy did not rank the strategies in the same way. GDV was more often near the oracle layer than adequacy-based selection, yet adequacy-based selection had much lower adequacy regret. This is why exact layer agreement is not the main criterion: layers far apart in

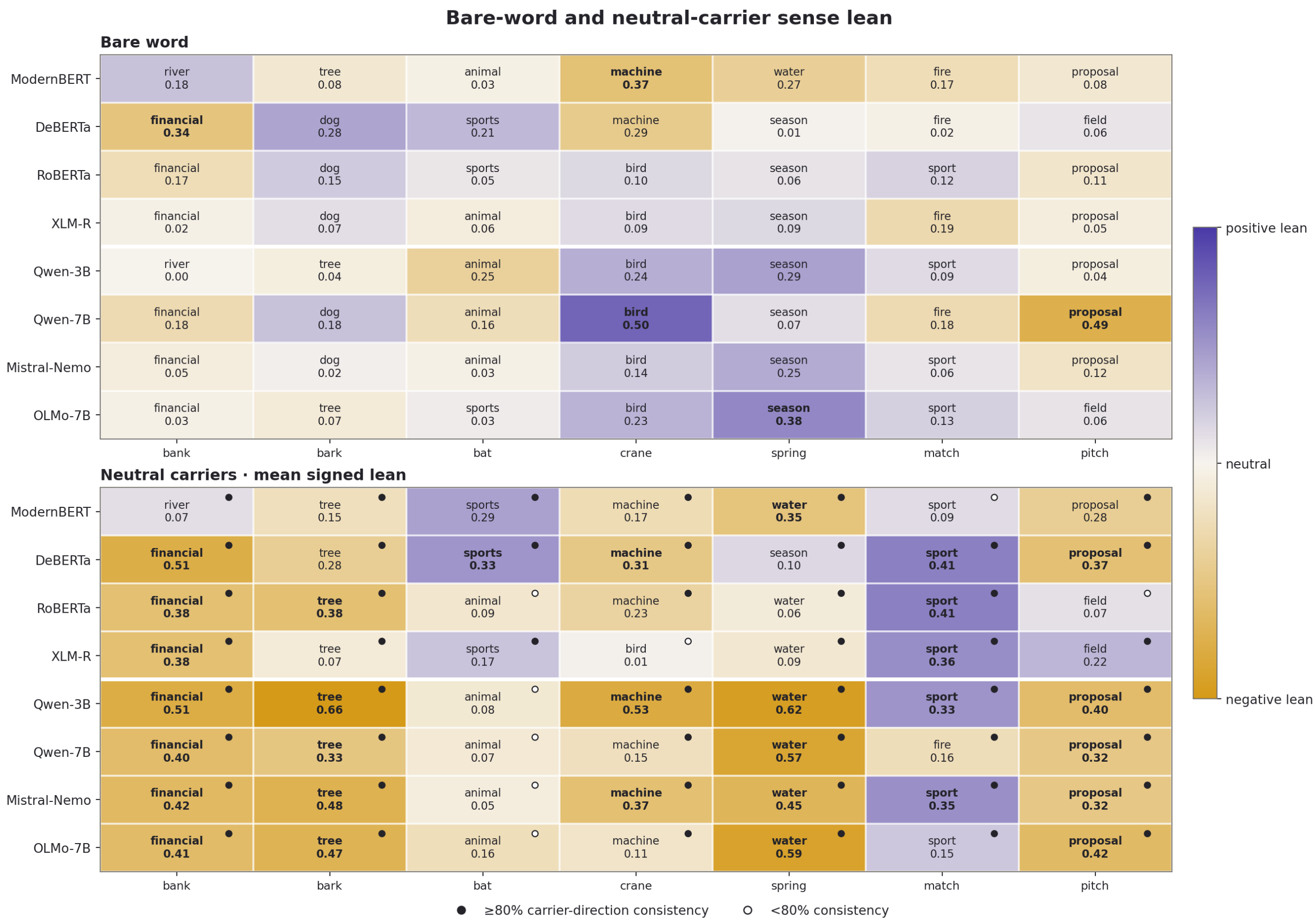

Fig. 1. Baseline sense lean for each model and homonym without explicitly disambiguating context. The upper panel shows the bare-word input, while the lower panel shows the mean lean across the five neutral carrier sentences. Each cell names the closer sense, and colour saturation represents the absolute strength of the normalised centroid-proximity lean. In the carrier panel, a filled dot indicates that at least four of the five carriers lean in the same direction. Repeated sense labels across models therefore identify homonyms for which nominally neutral inputs occupy a stable, asymmetric position in the two-sense geometry. The carriers associated with the strongest across-model asymmetries are listed in Table VII.

depth can support similar nearest-centroid decisions, while adjacent layers need not.

The Qwen2.5-7B *bat* example makes the measurement mismatch concrete (Figures 4 and 5). Layer 26 has a slightly more negative, nominally better GDV than layer 3 (-0.0634 versus -0.0505), but its full-space leave-one-out adequacy is only 0.60, compared with 0.90 at layer 3. This is not a mathematical contradiction. GDV compares average within- and between-class distances after per-feature standardisation; the adequacy margin evaluates each sentence's nearest centroid in the original hidden-state geometry. The example therefore shows that the two criteria can prefer different geometry.

D. H3: causal availability at the homonym

Encoder margins were almost unchanged when the clause moved from left to right of the homonym: mean normalised margin was 0.360 in L and 0.358 in R, giving a paired L–R effect of 0.0026. Decoder margins changed from 0.342 in L to approximately zero in R, giving an effect of 0.3420. The decoder-minus-encoder interaction was 0.3394, with a 95% crossed bootstrap interval of $[0.2358, 0.4355]$ (Table X).

The decoder R adequacy of 0.500 confirms the construction-based manipulation check. The inferential result is the paired L–R margin loss and its architecture interaction.

The architecture-level contrast was also consistent at the model level (Figure 6). Moving the decisive clause after the homonym had little effect in each encoder, whereas all four causal decoders showed a substantial reduction in correct-sense margin.

The decoder sensitivity was present for every homonym but varied in magnitude. The largest decoder-minus-encoder interactions occurred for *crane* (0.492), *match* (0.376), and *bat* (0.357), while *spring* (0.161) showed the weakest effects. Encoder left-minus-right changes remained close to zero for most words, with *match* showing the largest encoder change at 0.076. Thus the architecture interaction is global across words, while the strength of dependence on left context remains sense-

TABLE VII

STIMULUS AUDIT FOR THE FOUR HOMONYMS WITH THE STRONGEST ABSOLUTE ACROSS-MODEL MEAN CARRIER LEAN. MEAN B_l AVERAGES THE 40 CARRIER OBSERVATIONS FOR EACH WORD (8 MODELS $\times$ 5 CARRIERS); POSITIVE VALUES INDICATE SENSE 0 AND NEGATIVE VALUES SENSE 1. THE SEMANTIC DIRECTION IS STATED DIRECTLY TO MAKE THE SIGN INTERPRETABLE. LISTING THE ORIGINAL SENTENCES ALLOWS READERS TO JUDGE WHETHER THE NOMINALLY NEUTRAL CARRIERS MAY NEVERTHELESS FAVOUR ONE INTERPRETATION.

Word	Mean direction	Mean B_l	Neutral carriers
bank	Financial	-0.367	The bank was unstable.
			They examined the bank.
			The bank had changed.
			The bank needed protection.
			The bank became difficult to access.
bark	Tree	-0.350	The bark drew attention.
			They studied the bark.
			The bark was unusual.
			The bark changed noticeably.
			The bark became the focus.
spring	Water	-0.329	The spring was unusually dry.
			They monitored the spring.
			The spring changed quickly.
			The spring became a concern.
			The spring was hard to assess.
match	Sport	$+0.242$	The match was difficult to start.
			The match ended quickly.
			They checked the match.
			The match caused trouble.
			The match was watched closely.

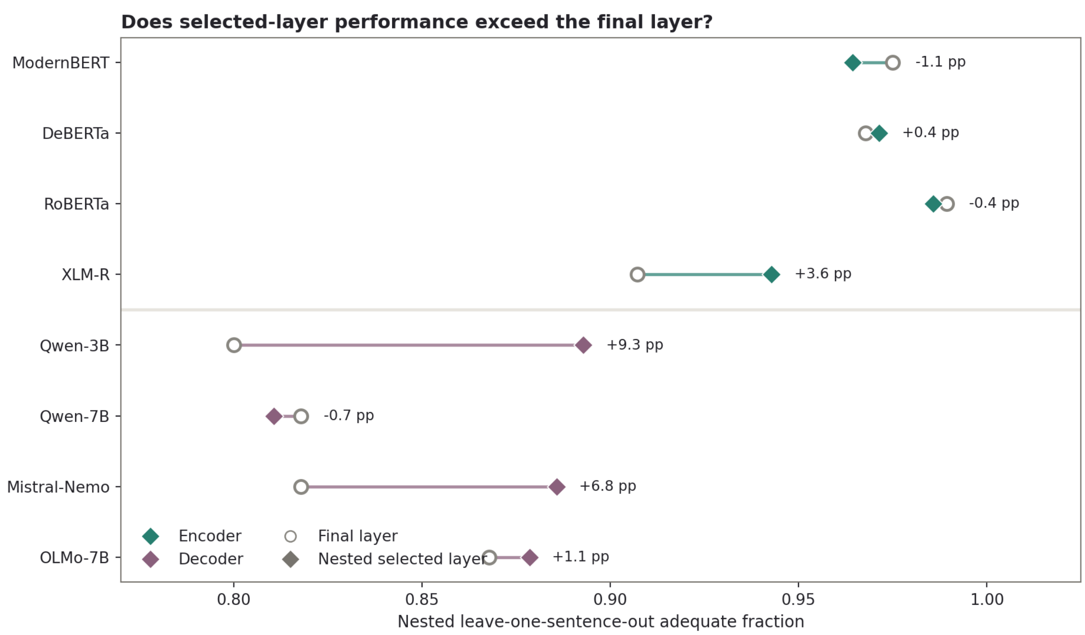

Fig. 2. Model-level comparison of nested selected-layer and final-layer H1 adequacy. Each estimate is the held-out adequate fraction averaged across the seven homonyms and weighted by the number of outer folds. Open circles show performance at the final layer, while diamonds show performance at the layer selected using only the corresponding training folds. The connecting segment shows the direction and magnitude of the difference: movement to the right indicates that layer selection improved held-out sense classification. Labels give the selected-minus-final difference in percentage points. Colour distinguishes the sampled encoders and decoders, which are separated by the horizontal divider.

and stimulus-specific.

E. H4: local decodability across sequence positions

As constructed, causal-decoder adequacy at the homonym was 0.500 because the two opposing R-condition sentences shared the same prefix at that position. At the sentence-final period, after the resolving clause had become causally available, adequacy increased to 0.761. Among decoder observations that were inadequate at the homonym, 67.9% became adequate at the endpoint; among those that were adequate at the homonym, 15.7% were no longer adequate there. The corresponding encoder comparison moved from 0.861 adequacy at the homonym to 0.668 at the endpoint, with a gain rate of 41.0% and a loss rate of 29.0% (Table XI).

The architecture-level direction was consistent across words.

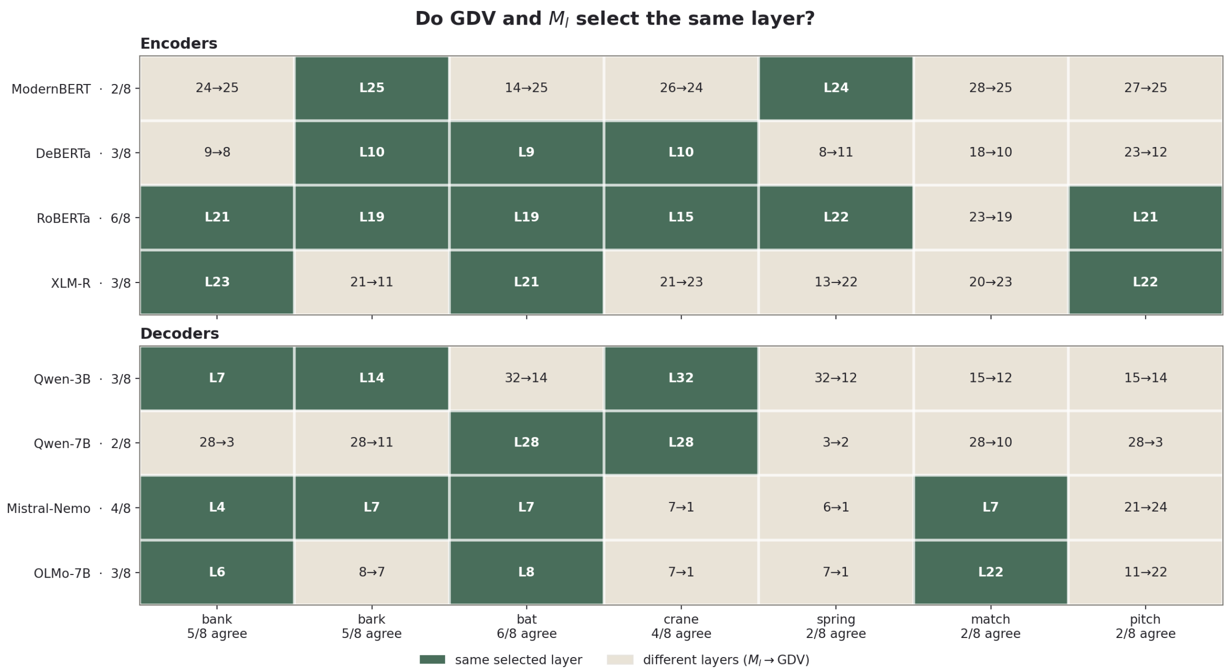

Fig. 3. Agreement between margin-based and GDV-based layer localisation for each model and homonym. The upper and lower panels show encoders and decoders, respectively. Agreement-coloured cells indicate that the full-profile nearest-centroid margin, M_l, and GDV select the same layer; contrasting cells indicate different selections. An agreeing cell prints the shared layer (e.g., L12), while a disagreeing cell prints the margin-selected layer followed by the GDV-selected layer (e.g., $12 \to 20$). Row and column annotations summarise agreement across words and models. The methods select exactly the same layer in 26 of 56 model–word combinations, showing that semantic-layer localisation depends substantially on the geometric criterion. Exact agreement is interpreted alongside held-out adequacy and oracle regret because different layers can nevertheless support similar decision performance.

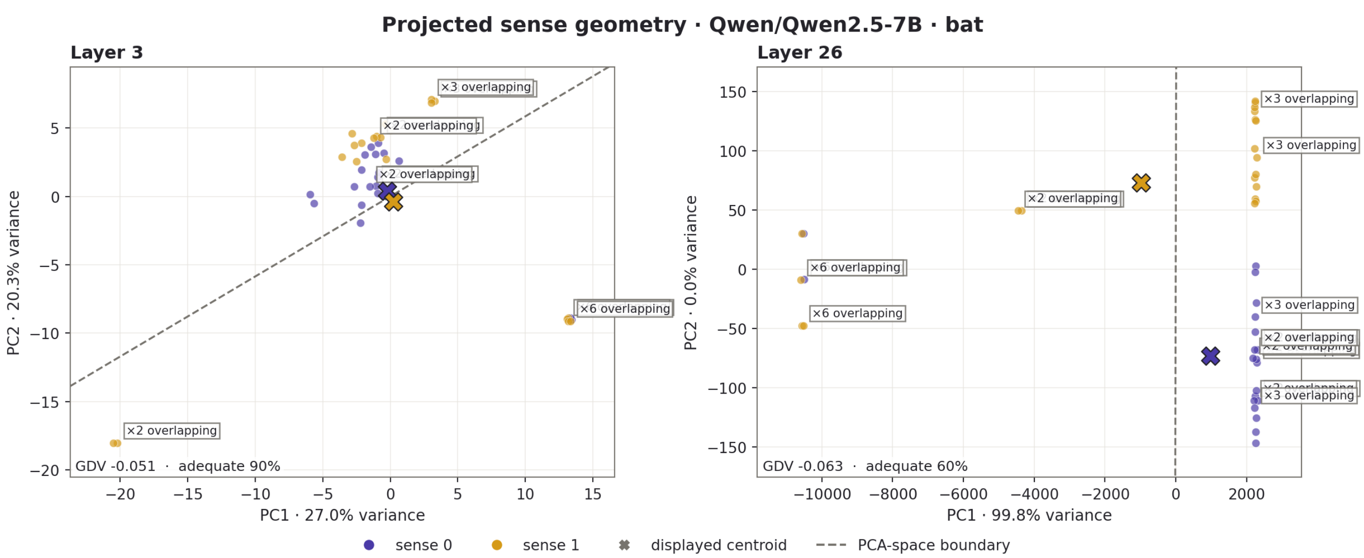

Fig. 4. Two-dimensional PCA projections of the Qwen2.5-7B representations for the two *bat* senses at layers 3 and 26. Points are profiling sentences, colours identify the authored senses, crosses mark the centroids of the displayed points, and the dashed line is the nearest-centroid decision boundary computed from the two plotted dimensions only, not the full-space classifier used for the reported adequacy values. Layer 26 has the more negative, nominally better GDV, whereas layer 3 supports higher leave-one-out adequacy. The projection illustrates how the sense geometry changes between the selected layers; all reported GDV and adequacy values are calculated in the original full-dimensional hidden space.

Decoder endpoint adequacy exceeded the constructed 0.500 homonym baseline for every homonym, although the magnitude of recovery varied substantially. *Crane* showed the highest conditional gain (0.950), followed by *pitch* (0.800) and *bank* and *bark* (0.750), whereas *spring* gained in only 0.200 of initially inadequate cases and reached final adequacy of 0.550.

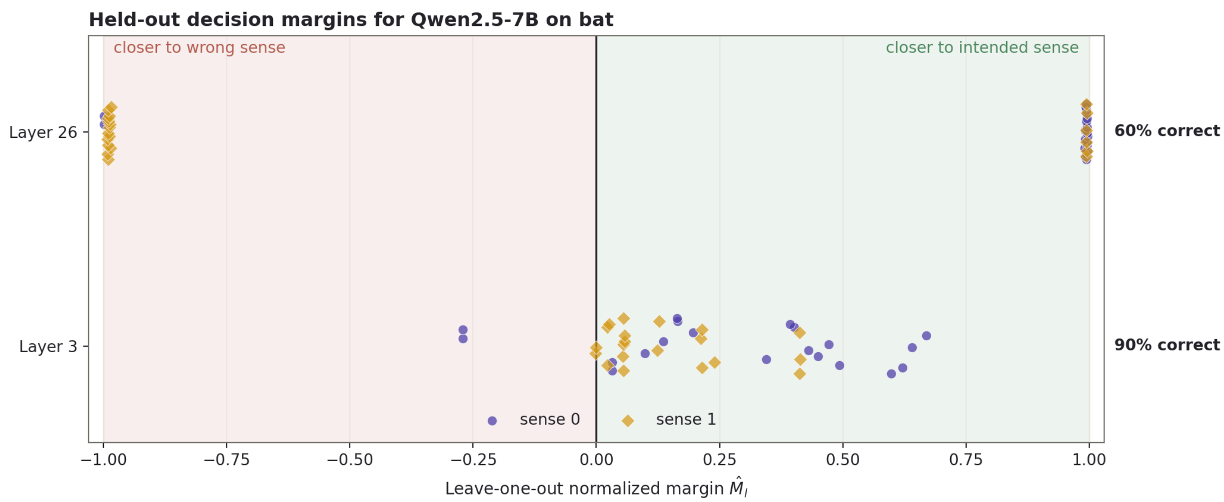

Fig. 5. Full-space leave-one-out decision scores for Qwen2.5-7B on *bat* at layers 3 and 26. Each point represents one profiling sentence scored against sense centroids estimated without that sentence. Positive normalised margins fall on the correct side of the nearest-centroid boundary, while negative margins indicate incorrect sense assignment; zero marks the boundary. Layer 3 correctly assigns 90% of the sentences, compared with 60% at layer 26, despite layer 26 having the more favourable GDV. The distributions therefore show at the observation level why stronger average class geometry does not necessarily produce better nearest-centroid decisions.

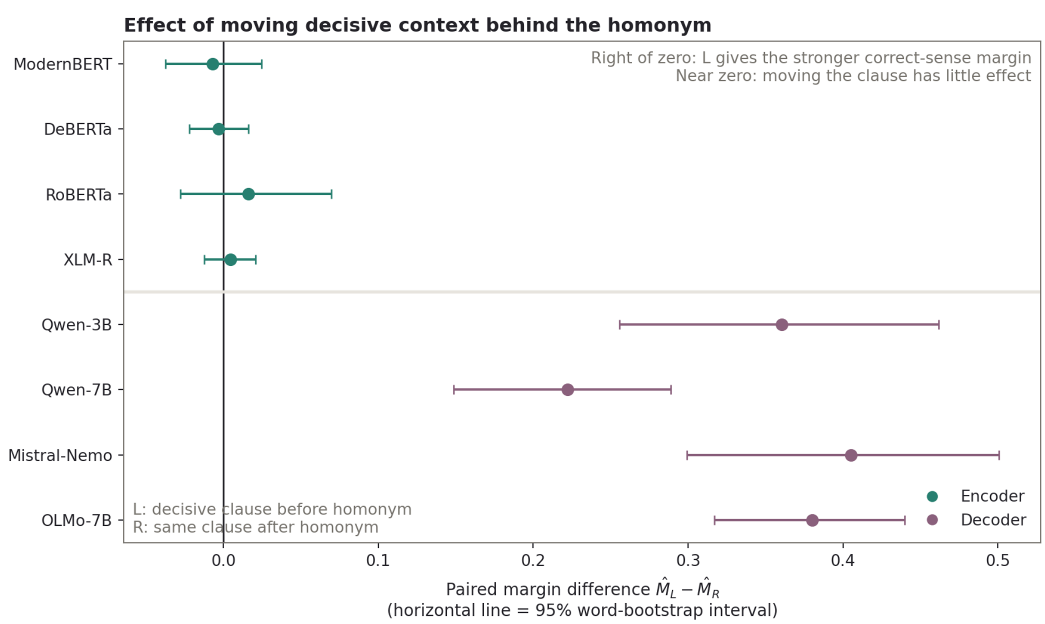

Fig. 6. Per-model effect of moving the decisive clause from before the homonym (L) to after it (R). Each point is the mean paired difference $\widehat{M}_L - \widehat{M}_R$ across homonyms; the horizontal interval is a 95% word-bootstrap interval. Values near zero mean that clause order changes little. Positive values mean that placing the decisive context before the homonym produces the stronger correct-sense margin. The large positive decoder effects reflect the causal mask; decoder R adequacy of 0.500 is a construction-based manipulation check rather than an empirical advantage.

Encoder endpoint adequacy was lower than homonym adequacy for every word; *crane* and *pitch*, for example, were already almost completely adequate at the homonym but were less decodable at the endpoint. The later-position benefit was therefore broad for decoders but strongly word-dependent in magnitude.

The diagnostic cross-position score—period states evaluated against homonym centroids—was near chance (0.516 overall), reinforcing that the two positions do not share a directly transferable centroid geometry.

F. H5: fixed-sentinel incremental updating

The balanced design produced 784 model–item observations: eight models, seven homonyms, and 14 items per homonym. The fixed sentinel leaned toward the primed, incorrect sense before the homonym ($\widehat{M} = -0.080$) and after it (-0.129). After the annotated resolver, the overall mean was approximately at the centroid boundary (-0.001). The primary

TABLE VIII
HELD-OUT H1 SENSE ADEQUACY UNDER NESTED LAYER SELECTION. EACH PROFILING SENTENCE FORMS ONE OUTER TEST FOLD; THE SELECTED LAYER IS CHOSEN USING ONLY THE REMAINING SENTENCES AND IS THEN EVALUATED ON THE UNTOUCHED SENTENCE. SELECTED AND LAST REPORT THE PROPORTION OF TEST SENTENCES ASSIGNED TO THE CORRECT SENSE AT THE SELECTED AND FINAL LAYERS, RESPECTIVELY. DIFFERENCE IS SELECTED MINUS LAST, SO POSITIVE VALUES INDICATE THAT INTERMEDIATE-LAYER SELECTION IMPROVES GENERALISATION OVER THE FINAL-LAYER DEFAULT. THE *Outer folds* COLUMN GIVES THE TOTAL NUMBER OF HELD-OUT EVALUATIONS.

Architecture	Outer folds	Selected	Last	Difference
Encoder	1,120	0.966	0.960	0.006
Decoder	1,120	0.867	0.826	0.041
All models	2,240	0.917	0.893	0.024

TABLE IX
CROSS-WORD H2 LAYER-SELECTION PERFORMANCE ACROSS 56 HELD-OUT MODEL–WORD FOLDS. ADEQUACY IS THE PROPORTION CORRECTLY ASSIGNED. ORACLE REGRET IS THE ADEQUACY LOST RELATIVE TO THE HELD-OUT-WORD ORACLE. LAYER DISTANCE IS THE ABSOLUTE DISTANCE FROM THE ORACLE LAYER, AND WITHIN 2 IS THE PROPORTION SELECTED WITHIN TWO LAYERS OF IT.

Strategy	Adequacy	Oracle regret	Layer distance	Within 2 layers
GDV	0.892	0.038	4.62	0.643
Adequacy-based cross-word	0.914	0.016	6.62	0.339
Last layer	0.893	0.037	11.96	0.250
Oracle	0.930	0.000	0.00	1.000

paired resolver-minus-homonym change was therefore $+0.128$; 605/784 observations (77.2%) moved toward the authored correct sense. A positive continuous change does not require a boundary crossing. Among the 580 observations whose homonym-stage sentinel was on the primed side, 234 (40.3%) reached the correct side by the resolver. The architecture-specific trajectories are shown in Table XII and Figure 7.

Both sampled architecture groups moved toward the correct sense on average. The decoder mean shift was larger, but its resolver-stage endpoint remained slightly on the primed side and its conditional transition rate was lower. This difference follows partly from the stronger decoder-side lean at the homonym: more decoder observations entered the conditional denominator, and they began farther from the boundary. These are descriptive patterns in the selected model sample rather than an isolated causal effect of architecture. A post-hoc crossed model–word bootstrap found no clear architecture difference in update magnitude: the decoder-minus-encoder difference was 0.018, with a 95% interval of $[-0.048, 0.080]$. The decoder resolver endpoint was lower by 0.043 ($[-0.087, -0.001]$), and its conflict-minus-matched control difference was more negative by 0.084 ($[-0.165, -0.010]$). These exploratory intervals distinguish movement from endpoint quality, but do not isolate bidirectional attention from the other differences between the model groups.

The aggregate architecture pattern was not uniform across homonyms. Decoder updates were larger for five of the seven words, but encoders moved farther for *bark* and especially *bat*.

TABLE X
EFFECT OF CAUSAL CONTEXT POSITION ON SENSE INFORMATION AT THE HOMONYM. $\widehat{M}_L$ AND $\widehat{M}_R$ ARE MEAN NORMALISED CORRECT-SENSE MARGINS WHEN THE DISAMBIGUATING CLAUSE APPEARS BEFORE (L) OR AFTER (R) THE HOMONYM; POSITIVE VALUES INDICATE PROXIMITY TO THE CORRECT SENSE CENTROID AND ZERO IS THE DECISION BOUNDARY. ADEQUACY L AND R GIVE THE CORRESPONDING PROPORTIONS OF OBSERVATIONS ON THE CORRECT SIDE OF THAT BOUNDARY. A LARGER L-MINUS-R DIFFERENCE INDICATES GREATER DEPENDENCE ON THE DISAMBIGUATING CONTEXT BEING CAUSALLY AVAILABLE WHEN THE HOMONYM IS PROCESSED.

Architecture	$\widehat{M}_L$	$\widehat{M}_R$	Adequacy L	Adequacy R
Encoder	0.360	0.358	0.843	0.861
Decoder	0.342	-0.0002	0.918	0.500

TABLE XI
CHANGE IN LOCAL SENSE DECODABILITY FROM THE HOMONYM TO THE SENTENCE-FINAL PERIOD. TARGET AND FINAL ARE THE PROPORTIONS ASSIGNED TO THE CORRECT SENSE USING POSITION-SPECIFIC CENTROIDS. GAIN IS THE PROPORTION OF TARGET-INADEQUATE OBSERVATIONS THAT BECOME ADEQUATE AT THE ENDPOINT; LOSS IS THE PROPORTION OF TARGET-ADEQUATE OBSERVATIONS THAT BECOME INADEQUATE THERE. FRACTIONS SHOW THE CONTRIBUTING COUNTS AND MAKE THE CONDITIONAL DENOMINATORS EXPLICIT.

Architecture	Target	Final	Gain	Loss
Encoder	0.861	0.668	16/39 = 0.410	70/241 = 0.290
Decoder	0.500	0.761	95/140 = 0.679	22/140 = 0.157

Encoder resolver endpoints were more correctly aligned for most words, whereas the decoder endpoint was higher for *pitch*. The architecture averages therefore describe a broad tendency toward stronger decoder priming and less complete endpoints, not an invariant word-level ordering.

At model level, the mean update was positive for all eight models and ranged from 0.074 for ModernBERT to 0.186 for DeBERTa. At the finer model–word level, 54/56 cells had a positive mean update; the two exceptions were XLM-R for *crane* (-0.009) and *spring* (-0.047).

Word-level heterogeneity is summarised in Table XIII.

Every homonym had a positive mean update, ranging from 0.068 for *spring* to 0.181 for *bank*. The conditional transition rate was more variable: *crane* and *spring* were close to 31%, whereas *pitch* reached 56.6%. The balanced directions were also asymmetric. For sense-0-to-sense-1 items, 125/256 initially primed observations crossed to the correct side (48.8%); for sense-1-to-sense-0 items, the corresponding rate was 109/324 (33.6%). Equal item counts therefore remove direction as a sampling imbalance, but do not make the two sense transitions equally easy or equally likely to be primed initially.

Several recurring word patterns are visible. *Spring* produced the smallest mean update, the lowest decoder endpoint adequacy in H4, and one of the lowest H5 transition rates. *Crane* showed strong recovery at the later H4 position but the largest H5 conflict cost, suggesting that access to the resolving context and removal of its earlier influence are distinct. *Bank* combined one of the strongest H0 baseline asymmetries with the largest H5 update, showing that a strong starting lean does not necessarily prevent movement.

TABLE XII

H5 UPDATING BY SAMPLED ARCHITECTURE. PRIME, HOMONYM, AND RESOLVER ARE MEAN NORMALISED MARGINS TOWARD THE AUTHORED RESOLVED SENSE AT THE THREE INCREMENTAL STAGES; NEGATIVE VALUES INDICATE THAT THE FIXED SENTINEL REMAINS CLOSER TO THE INITIALLY PRIMED, INCORRECT SENSE, WHILE POSITIVE VALUES PLACE IT CLOSER TO THE EVENTUAL CORRECT SENSE. Δ IS THE MEAN RESOLVER-MINUS-HOMONYM CHANGE, SO POSITIVE VALUES INDICATE UPDATING TOWARD THE RESOLUTION. POSITIVE Δ REPORTS HOW MANY OF THE 392 OBSERVATIONS PER ARCHITECTURE MOVED IN THAT DIRECTION. TRANSITION REPORTS THE PROPORTION THAT CROSSED FROM THE PRIMED SIDE AT THE HOMONYM TO THE CORRECT SIDE AFTER THE RESOLVER; ITS DENOMINATOR INCLUDES ONLY OBSERVATIONS THAT WERE INITIALLY PRIMED. THE TABLE DISTINGUISHES MOVEMENT TOWARD THE CORRECT SENSE FROM COMPLETE BOUNDARY CROSSING.

Arch.	Prime	Homonym	Resolver	Δ	Positive Δ	Transition
Encoder	-0.062	-0.098	0.021	0.119	284/392 = 72.4%	111/262 = 42.4%
Decoder	-0.098	-0.159	-0.022	0.137	321/392 = 81.9%	123/318 = 38.7%

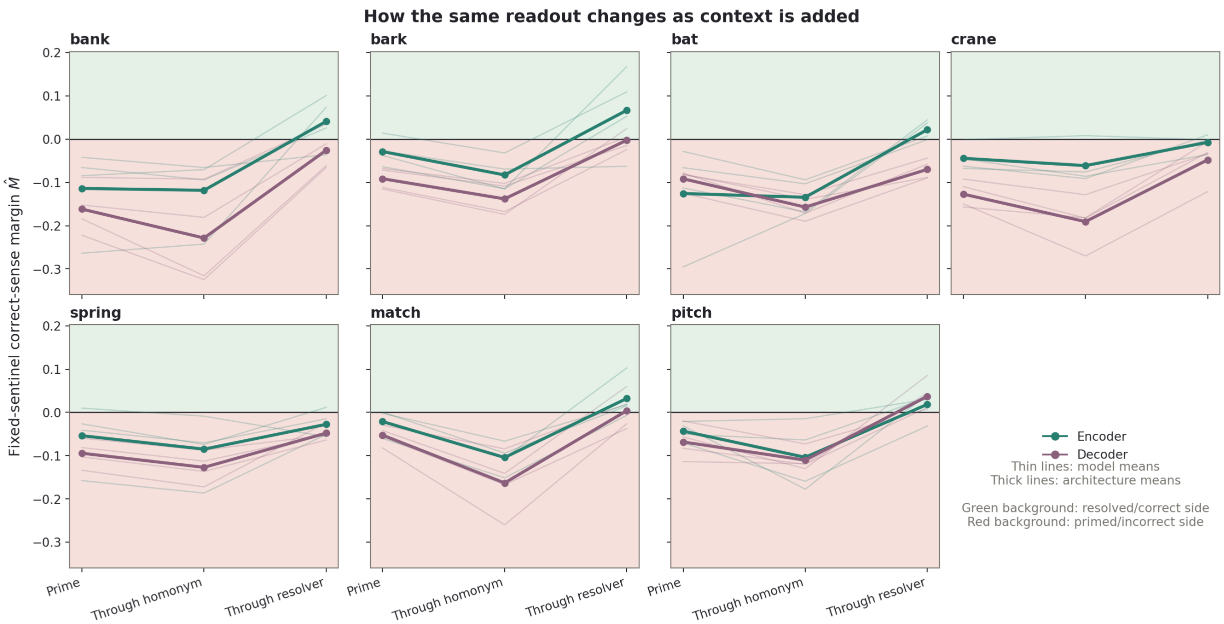

Fig. 7. Trajectory of the fixed-sentinel margin as progressively more context is supplied for each homonym. Prime, Homonym, and Resolver correspond to reruns ending before the homonym, at the homonym, and at the annotated resolver. Every point reads the same appended token, `probe`, at the same selected layer. Positive margins place the sentinel closer to the authored resolved-sense centroid, while negative margins place it closer to the initially primed, incorrect sense; zero is the centroid decision boundary. Thin lines show individual model means across the 14 items for that homonym, and thick lines show encoder and decoder means. Resolving context generally moves the readout toward the correct sense, but starting position, update magnitude, and final endpoint vary substantially. Green and red backgrounds mark the resolved/correct and primed/incorrect sides of the sense axis.

The resolver-stage conflict margin remained below the matched non-conflicting control by 0.140 on average. The difference was -0.098 for encoders and -0.182 for decoders, and it was negative in 55/56 model–word combinations. Thus, even with the same homonym and resolver, preceding conflicting context was associated with a lower correct-sense endpoint. The conflict endpoint was also 0.056 below the resolver-in-isolation baseline overall (encoder -0.018, decoder -0.093), but this comparison was less uniform: 45/56 cells were negative. The resolver itself therefore supplies substantial sense-diagnostic information, while the full preceding context modulates how strongly that information appears at the fixed sentinel. These paired comparisons are visualised in Figure 8. Categorical endpoint summaries depended strongly on the descriptive margin threshold. With a zero threshold, 404/784 resolver-stage states (51.5%) were on the correct side. At $|\widehat{M}| > 0.1$, 244 (31.1%) were classified as correct, 247 (31.5%) as primed, and 293 (37.4%) remained in the boundary region. At $|\widehat{M}| > 0.3$, 666 (84.9%) were in the boundary region, with only 61 (7.8%) correct and 57 (7.3%) primed. This sensitivity is why the continuous paired change, rather than a selected confidence threshold, is the primary H5 result.

Together, the continuous shifts and matched-control differences show broad updating alongside a persistent effect of preceding conflict.

G. Exploratory relationships across hypotheses

A post-hoc analysis tested whether H0 baseline lean was related to the later experiments. Across all 56 model–word combinations, absolute carrier lean had little association with nested H1 selected-layer adequacy (Spearman $\rho = -0.143$). For H5, the signed carrier lean was aligned with each primed direction before aggregation. Stronger alignment with the primed sense was associated with a more negative correct-sense margin both at the homonym ($\rho = -0.227$; 112 direction cells) and at the resolver ($\rho = -0.251$; 112 cells). In contrast,

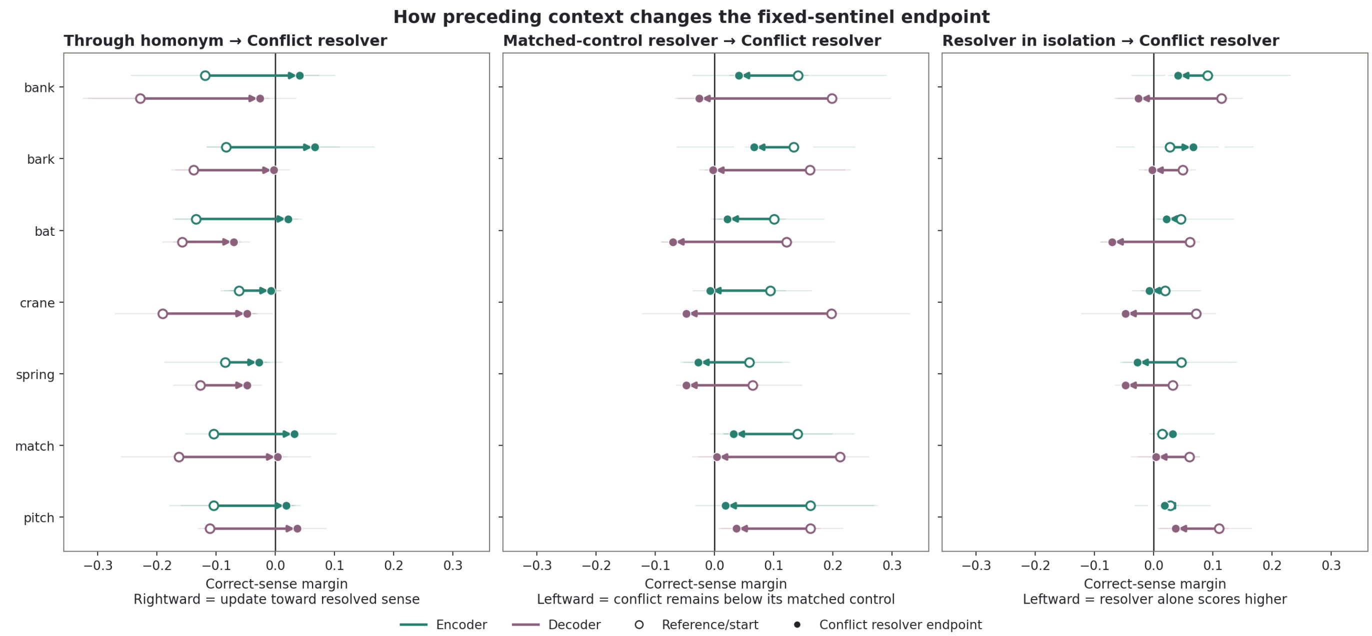

Fig. 8. Three paired H5 comparisons on the fixed sentinel's normalised correct-sense margin. Open circles mark the reference state, filled circles mark the conflict-item resolver endpoint, and arrows show the paired difference. Movement to the right indicates stronger correct-sense alignment. The left panel measures updating from the conflict homonym prefix to its resolver prefix. The middle panel begins at the matched non-conflicting resolver endpoint, so movement to the left shows the remaining cost of preceding conflict. The right panel begins with the resolver in isolation and shows how the full conflict context changes its effect. Thin arrows represent individual models for each homonym; thick arrows show the corresponding encoder and decoder means.

TABLE XIII
WORD-LEVEL H5 OUTCOMES. MEAN Δ IS THE RESOLVER-MINUS-HOMONYM CHANGE. POSITIVE Δ IS THE PROPORTION MOVING TOWARD THE RESOLVED SENSE. TRANSITION IS THE BOUNDARY-CROSSING RATE AMONG INITIALLY PRIMED OBSERVATIONS. CONFLICT–CONTROL IS THE RESOLVER-STAGE CONFLICT MARGIN MINUS THE MATCHED-CONTROL MARGIN.

Word	Mean Δ	Positive Δ	Transition	Conflict–control
bank	0.181	93/112 (83.0%)	38/90 (42.2%)	−0.162
bark	0.143	101/112 (90.2%)	35/83 (42.2%)	−0.116
bat	0.122	81/112 (72.3%)	31/78 (39.7%)	−0.135
crane	0.098	80/112 (71.4%)	27/87 (31.0%)	−0.174
spring	0.068	79/112 (70.5%)	24/77 (31.2%)	−0.099
match	0.152	97/112 (86.6%)	32/82 (39.0%)	−0.158
pitch	0.135	74/112 (66.1%)	47/83 (56.6%)	−0.135

it was essentially unrelated to the resolver-minus-homonym update magnitude ($\rho = -0.026$) and only weakly related to conditional transition success ($\rho = -0.146$; 107 cells with a non-zero primed denominator).

The exploratory pattern is consistent with a persistent baseline offset: stronger H0 alignment with the primed sense corresponded to more primed positions at both the homonym and resolver, but not to a smaller or larger resolver-induced movement. The word-level results support the same distinction. For example, *bank* combined a strong baseline lean with the largest H5 update, while *spring* combined a strong lean with the smallest update. These post-hoc relationships describe recurring patterns in the present sample and require confirmatory hierarchical testing.

IV. DISCUSSION

This study tested whether contextual semantic information becomes most strongly expressed within a reproducible region of transformer depth. The results provide little support for that simple layer–stage correspondence. Homonym senses were widely distinguishable, and intermediate layers were useful in some models, but the preferred depth varied across models, words, and geometric criteria. Across the six analyses, semantic accessibility was readout-dependent: it varied with the metric, layer, token position, available context, and preceding context.

A. Semantic localisation is criterion- and stimulus-dependent

H0 showed that the two-sense geometry was often asymmetric before explicitly disambiguating context was supplied. Some directions appeared compatible with an intuitive dominant sense, such as the financial lean of *bank*; others, including tree covering for *bark* and water source for *spring*, were less easily explained by assumed lexical frequency. Bare-word and carrier directions also frequently differed. The baseline should therefore be interpreted as a model–word–stimulus asymmetry within the profiling geometry, not as a direct estimate of sense frequency in pretraining. It may combine training-distribution effects with the lexical and syntactic affordances of the carriers, tokenisation, target position, category breadth, and cluster shape.

These unequal starting positions matter because contextual evidence acts on a representation that may already be displaced toward one sense. The exploratory H0–H5 analysis was consistent with a persistent offset: stronger alignment with the primed sense was associated with more primed positions at both the homonym and resolver, but not with a smaller or larger resolver-induced movement. Updating therefore appeared to begin from unequal starting positions rather than from uniformly different responsiveness. Because this relationship was identified post hoc, it motivates confirmatory analysis rather than a causal claim.

H1 and H2 further weakened the expectation of a stable semantic depth. Intermediate-layer selection improved held-out sense discrimination in some models, particularly among the sampled causal decoders, but the benefit did not generalise uniformly. More importantly, GDV-based and adequacy-based selection often preferred different layers or depth regions. A semantic stage that is stable only under one criterion is difficult to treat as a general component of an ordered processing hierarchy.

The disagreement is informative about the activation geometry. GDV evaluates standardised average within- and between-group distances, whereas adequacy evaluates held-out nearest-centroid decisions in the original hidden space. A layer can therefore improve global class organisation while moving a smaller set of difficult observations across the wrong boundary, or support reliable item-level decisions without producing the most compact global clusters. The Qwen2.5-7B *bat* example makes this distinction concrete: the layer with the more favourable GDV produced substantially worse sentence-level decisions. Layer number alone is consequently an incomplete description of semantic quality.

B. Incremental context limits the layer-stage analogy

H3 establishes a boundary on the interpretation of transformer depth. For a causal decoder, additional layers can transform only the context already available at the homonym; they cannot incorporate a disambiguating word that appears later. Encoders can use that right context immediately. The resulting clause-order interaction is expected from the attention masks, but it demonstrates why stronger separation at a later computational point cannot automatically be interpreted as deeper semantic processing.

H4 shows where the missing information can subsequently become accessible. At the sentence endpoint, causal decoders had encountered both the ambiguous word and its resolver, and many senses that were unavailable at the homonym became locally decodable. Semantic information was therefore not expressed exclusively at the ambiguous token. Claims that a model contains or lacks sense information must specify both the layer and the token position from which it is read.

Causal decoders resemble human comprehension only in the limited sense that later words are unavailable when an earlier word is first encountered. Human language processing combines incremental input with recurrent, predictive, bottom-up, and top-down interactions, whereas the decoder applies a fixed causal mask across transformer blocks. Encoders provide a useful full-context contrast but are likewise unlike ordinary incremental comprehension. H3 and H4 therefore separate computational access regimes; they do not identify either architecture as a direct model of the human language system. Their main lesson is that layer depth, reading time, and accumulated context cannot be treated as the same axis.

C. Semantic revision is path-dependent

H5 extended the analysis from semantic accessibility to semantic revision. Resolving information moved the fixed-sentinel readout toward the authored sense across all models and homonyms, but movement was more common than boundary crossing. The readout often remained near the centroid boundary, and complete geometric resolution—crossing from the primed to the resolved side of that boundary—occurred in a minority of initially primed observations.

The matched controls strengthen the interpretation beyond a general claim about weak resolvers. Conflict items and controls contained the same homonym and resolver, yet conflict endpoints remained less aligned with the resolved sense. Earlier semantic commitment therefore left a measurable trace after contradictory information arrived. The isolated-resolver comparison further showed that the resolver itself supplied a sense-diagnostic signal, while the preceding context affected how fully that signal was expressed. Updating and persistence coexisted within the same readout.

Persistence is not inherently undesirable: a language system must preserve earlier information while it remains valid. The functional issue is calibration when later evidence sharply contradicts the established interpretation. In H5, the resolver usually moved the representation in the appropriate direction but often did not overcome the earlier interpretation completely. The predominance of partial movement over boundary crossing is therefore evidence of contextual inertia under the present geometric proxy.

Human comprehenders can also retain traces of an initial garden-path interpretation, and reanalysis is not invariably complete [13]. Residual ambiguity is therefore not a categorical difference between humans and language models. Humans usually arrive at a usable interpretation despite temporary ambiguity; in the present model-internal proxy, however, complete geometric resolution occurred in a minority of initially primed observations. Because the study contains no matched human responses, those rates cannot be compared directly. The model result instead provides a measurable candidate for a future matched experiment.

The sampled encoders reached more correctly aligned endpoints and retained a smaller conflict–control difference than the sampled decoders, even though decoders sometimes moved farther because they began more strongly primed. This suggests differences in how prior and resolving evidence are integrated, but the heterogeneous model sample cannot attribute them uniquely to the attention mask. Outcomes also varied by homonym and direction. Equal item counts remove sampling imbalance, but they do not equalise prime plausibility, resolver diagnosticity, naturalness, baseline dominance, or profiling geometry.

D. What behavioural similarity licenses

Behavioural similarity licenses a search for functional analogues but underdetermines their internal organisation. A human and a transformer may both use context to resolve *bank* correctly while arriving at that output through different intermediate operations. The present results specifically challenge a simple mapping from transformer depth to a sequence of human-like processing stages: semantic information was distributed across depth, different geometric properties peaked at different layers, and causal availability constrained what any layer could express.

Functional correspondences may therefore be more useful at the level of computational roles—prediction, integration, conflict detection, and revision—than as one-to-one mappings onto layers or cortical regions. Violation-sensitive measures offer one possible bridge. Neural signals such as the N400 are informative because they vary when incoming language conflicts with semantic expectation, not because they identify one anatomical semantic location. The H5 conflict–control difference is not an N400 analogue, but it provides a model-side conflict measure that could be compared with human behavioural or neural responses in a matched experiment.

The study therefore redirects the original question. Rather than asking only where semantics lives, future comparisons should ask which model signals perform functions analogous to human prediction, conflict detection, and revision; how those signals evolve with incoming context; and whether they predict trial-level human behaviour or neural activity.

E. Limitations

The principal limitation is stimulus control. Holding the homonym constant removes lexical identity from the sense contrast, but the authored contexts still differ in vocabulary, topic, syntax, target position, and length. Profiling geometry can therefore reflect contextual regularities beyond the intended semantic distinction. The H0 carrier audit and layer-0 diagnostic make some dependencies visible, but context-only classification, target masking, counterbalanced templates, and lexically disjoint train–test sets are needed to quantify them directly.

The binary centroid axis is a deliberately restricted representation of lexical meaning. Several homonyms have additional senses, and the two authored regions may differ in breadth or internal variability. Margin adequacy and GDV measure accessibility within this imposed contrast rather than a word's complete semantic representation. Decodability also establishes only that labelled information is recoverable from a hidden state. Causal interventions, activation patching, and model-output measures are needed to determine whether the identified geometry contributes to behaviour.

The fixed sentinel stabilises token identity and terminal readout role across H5 prefixes but introduces an artificial continuation. Its absolute sequence position changes with prefix length, so a length-matched non-semantic prefix control would be required to isolate contextual updating completely from positional effects. Replication with multiple sentinels, pooled sentence representations, and behavioural outputs would show whether the path-dependent effect generalises beyond this instrument.

Finally, the model sample is small and heterogeneous. Encoder–decoder differences are entangled with training objective, parameter count, training data, tokenisation, and other architectural choices. Repeated observations within models, words, and items also create dependencies that are only partly captured by the present bootstrap analyses. The absence of human norms further limits the human comparison: ambiguity, prime strength, resolver clarity, reading cost, and final interpretation remain unknown for the same items. Architecture-level and human-comparative claims therefore require matched model families, hierarchical analyses, and direct human measurements.

F. Future work

The broad stage-order hypothesis should next be tested with parallel held-out diagnostics for surface form, syntax, contextual sense, task instruction, and output-relevant information. Counterbalanced contrasts would show whether these signals peak in a reproducible order, overlap across depth, or reorganise across model families. Layer-selection work should likewise focus on the stability of broader depth regions rather than exact agreement alone.

The H0–H5 relationship should be tested confirmatorily with a hierarchical model of baseline lean, prime direction, resolver strength, and update magnitude. Stronger stimulus controls would help distinguish geometric starting bias from properties of the authored sentences. H5 should also be extended to corrected entity attributes, changed event states, belief revisions, and repeated updates over longer contexts to test whether path dependence generalises beyond lexical reinterpretation.

Finally, a human comparison should use the same stimuli to measure initial interpretation, reading or listening cost, final comprehension, and lingering misinterpretation. Trial-level correspondence would show whether the model's conflict cost resembles a human revision signal in direction, degree, or only at a more abstract functional level.

G. Conclusion

The study finds accessible semantic structure without identifying one stable semantic stage. Sense-conditioned geometry varies with the word, model, selection criterion, token position, and contextual history. Depth alone therefore provides an incomplete account of where semantic information is expressed.

The clearest functional result concerns revision. Resolving context usually moves the representation toward the appropriate sense, but earlier conflicting context leaves a persistent cost and makes complete geometric resolution comparatively rare. Semantic processing in these models is therefore both context-sensitive and path-dependent.

Behavioural similarity remains a productive reason to compare humans and language models, but it does not entail a shared ordering of intermediate operations. The present results replace the search for one semantic layer with a more precise question: how semantic information becomes accessible, conflicted, and revised across different computational organisations.

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